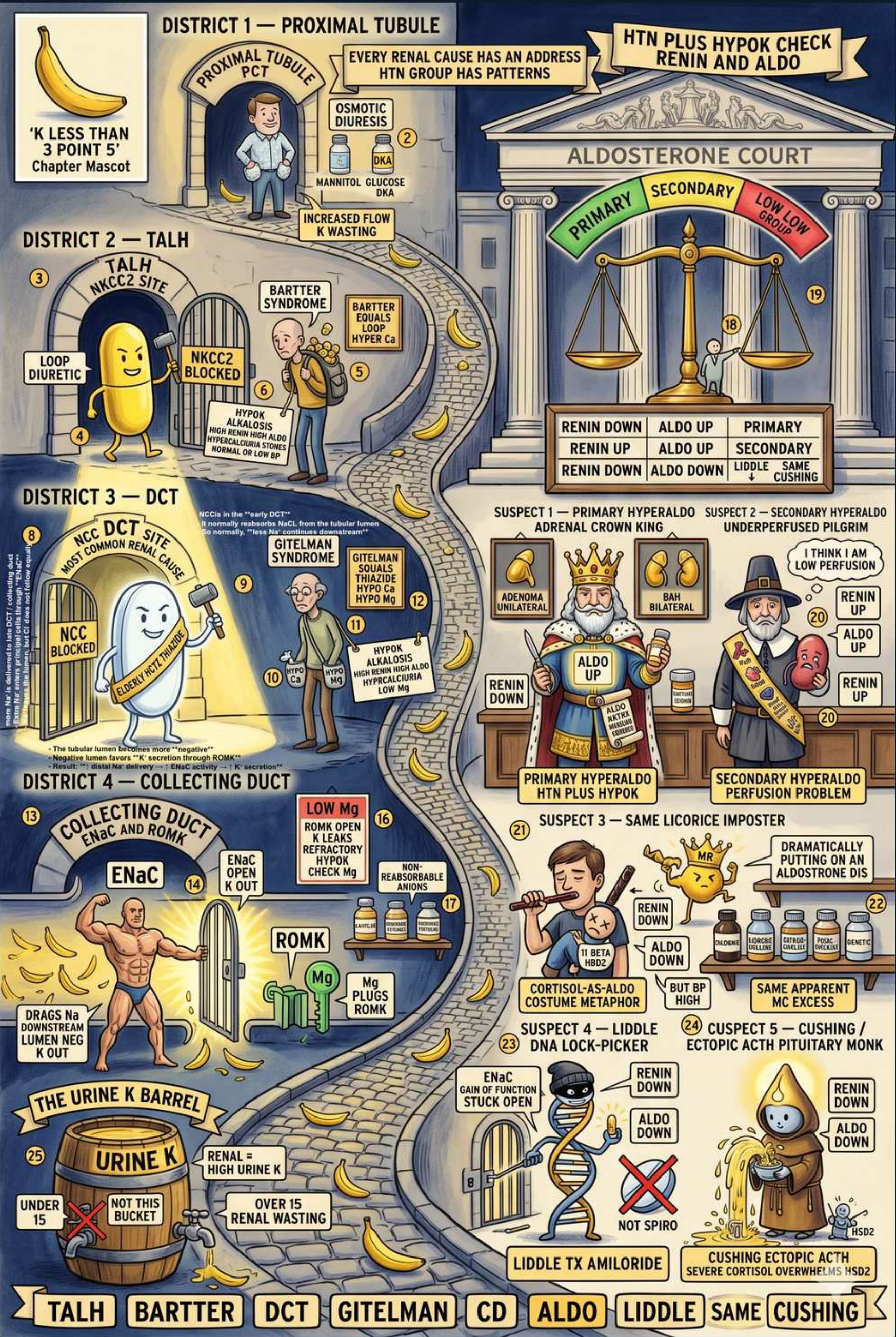


Hypokalemia — renal K loss



## 1. Renal loss = check renin/aldo + BP

### WHAT YOU SEE

Banner: HTN plus hypoK, check renin and aldo

### WHAT IT ENCODES

Renal K wasting (urine K >15 mmol/day, TTKG >4) is sorted first by BLOOD PRESSURE, then by renin/aldosterone. HTN + hypokalemia → mineralocorticoid-excess workup with an aldosterone:renin ratio. Normal BP → tubulopathy or diuretic.

### BOARD TRAP

Ordering renin/aldo in a NORMOTENSIVE hypokalemic patient. That picture is Bartter, Gitelman, or a diuretic — the renin/aldo workup only pays off once HTN is confirmed.

## 2. Proximal tubule / diuretics

### WHAT YOU SEE

District 1 — proximal tubule (PCT), mannitol/glucose diuresis, increased flow K wasting

### WHAT IT ENCODES

Osmotic diuresis from the proximal tubule — glucose (DKA/HHS), mannitol, urea — increases distal tubular flow, which drives flow-dependent K secretion through ROMK independent of aldosterone. BP is normal. Acetazolamide (proximal/type-2 RTA-like) also wastes K.

### BOARD TRAP

Forgetting that the flow effect alone causes K loss without high aldosterone — a normotensive hyperglycemic patient is wasting K via flow, so chasing an adrenal cause is misdirected.

## 3. Loop diuretic / TALH / Bartter

### WHAT YOU SEE

District 2 — TALH, NKCC2 site, loop diuretic and Bartter both block NKCC2

### WHAT IT ENCODES

In the thick ascending limb, loop diuretics (furosemide) and Bartter syndrome both block the NKCC2 cotransporter → hypokalemic metabolic alkalosis with NORMAL BP and HYPERcalciuria (nephrocalcinosis). Bartter is the genetic mimic of chronic furosemide use.

### BOARD TRAP

Diagnosing Bartter without first screening the urine for diuretics. Surreptitious loop-diuretic abuse reproduces the exact electrolyte picture — a urine diuretic screen must be negative before committing to Bartter.

## 4. DCT / Gitelman / thiazide

### WHAT YOU SEE

District 3 — DCT, NCC blocked, Gitelman = thiazide costume, low Mg low Ca urine

### WHAT IT ENCODES

In the distal convoluted tubule, Gitelman syndrome blocks the NCC cotransporter just like a THIAZIDE → hypokalemic metabolic alkalosis, NORMAL BP, LOW serum Mg, and LOW urine Ca (hypocalciuria). It usually presents later/milder than Bartter.

### BOARD TRAP

Confusing Gitelman with Bartter. Gitelman = thiazide-like = HYPOcalciuria + HYPOmagnesemia; Bartter = loop-like = HYPERcalciuria. The urine calcium is the discriminator, and the hypomagnesemia of Gitelman must be corrected to fix the K.

## 5. Collecting duct / ENaC

### WHAT YOU SEE

District 4 — collecting duct, ENaC + ROMK, Na in / K out

### WHAT IT ENCODES

In the principal cells of the collecting duct, Na reabsorption through ENaC creates a lumen-negative voltage that drives K secretion through ROMK. Aldosterone (and increased distal Na delivery) upregulates ENaC → accelerated K loss. This is the final common pathway for renal K wasting.

### BOARD TRAP

Missing that any condition increasing distal Na delivery (high-flow states, non-reabsorbable anions like bicarbonate or penicillins) augments K secretion here even without high aldosterone.

## 6. Mg plugs ROMK

### WHAT YOU SEE

Low Mg → ROMK leaks K, Mg plugs ROMK

### WHAT IT ENCODES

Intracellular magnesium normally blocks the ROMK channel from inside. Hypomagnesemia unplugs ROMK → unopposed K secretion → ongoing renal K wasting and REFRACTORY hypokalemia. Common with diarrhea, alcohol, PPIs, aminoglycosides, cisplatin, and Gitelman.

### BOARD TRAP

Repleting K without checking/correcting Mg. You CANNOT correct hypokalemia until magnesium is replaced — this is a guaranteed exam point and a real bedside failure mode.

## 7. Aldosterone court / renin-aldo

### WHAT YOU SEE

Aldosterone court balance: primary high aldo, secondary, low-low

### WHAT IT ENCODES

For HTN + hypokalemia the renin/aldo pattern names the cause: PRIMARY hyperaldosteronism = renin DOWN, aldo UP (ARR >20–30). SECONDARY = renin UP, aldo UP. NON-aldosterone mineralocorticoid effect (Liddle, licorice, Cushing) = renin DOWN, aldo DOWN.

### BOARD TRAP

Interpreting the aldo:renin ratio before HTN is established, or before stopping interfering drugs. ACE inhibitors/ARBs/spironolactone and diuretics distort renin and aldo — wash them out (or use amlodipine/doxazosin) before testing.

## 8. Primary hyperaldo (Conn)

### WHAT YOU SEE

Suspect 1 — adrenal crown king: aldo up, renin down, HTN plus hypoK, adenoma/bilateral

### WHAT IT ENCODES

Primary hyperaldosteronism (Conn) — aldosterone-producing adenoma or bilateral adrenal hyperplasia: HTN + hypokalemic metabolic alkalosis, ALDO high, RENIN suppressed. Confirm with saline-suppression/captopril test, then adrenal vein sampling to lateralize.

### BOARD TRAP

Treating bilateral hyperplasia surgically. Only a unilateral adenoma is resected; bilateral hyperplasia is managed medically with spironolactone or eplerenone — adrenal vein sampling, not CT alone, decides which.

## 9. Secondary hyperaldo

### WHAT YOU SEE

Suspect 2 — underperfused pilgrim: renin up, aldo up, perfusion problem (RAS)

### WHAT IT ENCODES

Secondary hyperaldosteronism: an underperfused kidney drives the RAAS → RENIN high → aldo high. Causes include renal artery stenosis (young woman = fibromuscular dysplasia; older = atherosclerosis), malignant HTN, and renin-secreting tumors.

### BOARD TRAP

Calling RAS 'Conn.' Both give HTN + hypoK + high aldo, but in secondary hyperaldo RENIN is HIGH (driving the system) whereas in Conn renin is SUPPRESSED — the renin level separates them.

## 10. Licorice / apparent MC excess

### WHAT YOU SEE

Suspect 3 — licorice imposter, cortisol-as-aldo costume, putting on aldosterone dress

### WHAT IT ENCODES

Licorice (glycyrrhizic acid) and congenital apparent mineralocorticoid excess inhibit renal  $11\beta$ -HSD2. Cortisol is then no longer inactivated to cortisone and floods the mineralocorticoid receptor → HTN + hypokalemic alkalosis with BOTH renin and aldosterone LOW.

### BOARD TRAP

Diagnosing Conn whenever HTN + hypoK appears. Low renin AND low aldo means a non-aldosterone mineralocorticoid effect (licorice, Liddle, Cushing/AME) — an aldosterone-secreting adenoma would have HIGH aldo.

## 11. Liddle (ENaC stuck open)

### WHAT YOU SEE

Suspect 4 — DNA lock-picker, ENaC costume stuck open, Liddle tx amiloride not spiro

### WHAT IT ENCODES

Liddle syndrome: autosomal-dominant gain-of-function mutation leaves ENaC constitutively OPEN → Na retention, HTN, hypokalemic alkalosis, with renin AND aldosterone LOW. Presents young with a family history.

### BOARD TRAP

Choosing spironolactone for Liddle. The defect is the channel itself, not aldosterone, so an MR antagonist fails — the answer is amiloride or triamterene, which block ENaC directly (plus low-sodium diet).

## 12. Cushing / ectopic ACTH

### WHAT YOU SEE

Suspect 5 — Cushing ectopic ACTH monk, severe cortisol overwhelms  $11\beta$ -HSD2

### WHAT IT ENCODES

Ectopic ACTH (e.g. small-cell lung cancer) or severe Cushing produces cortisol in quantities that overwhelm/saturate  $11\beta$ -HSD2 → cortisol spills onto the mineralocorticoid receptor → HTN + hypokalemic alkalosis with renin and aldo LOW.

### BOARD TRAP

Severe hypokalemic alkalosis + HTN + hyperpigmentation + rapid onset with very high ACTH points to ECTOPIC ACTH (paraneoplastic), not a slow-burning pituitary adenoma — the magnitude and tempo distinguish them.

## 13. Urine K barrel

### WHAT YOU SEE

The urine K barrel: under 15 = not the leak; over 15 = renal wasting

### WHAT IT ENCODES

Renal loss is confirmed when urine K >15 mmol/day (or TTKG >4, spot >13 mmol/g Cr). Only then do you split by BP and order renin/aldo. Low urine K sends you back to the GI/extrarenal or shift arms.

### BOARD TRAP

Calling it renal loss without verifying urine K is high. A low urine K means the kidney is conserving appropriately — the leak is GI/skin or the K is merely shifted, and a renal/adrenal workup would be wasted.